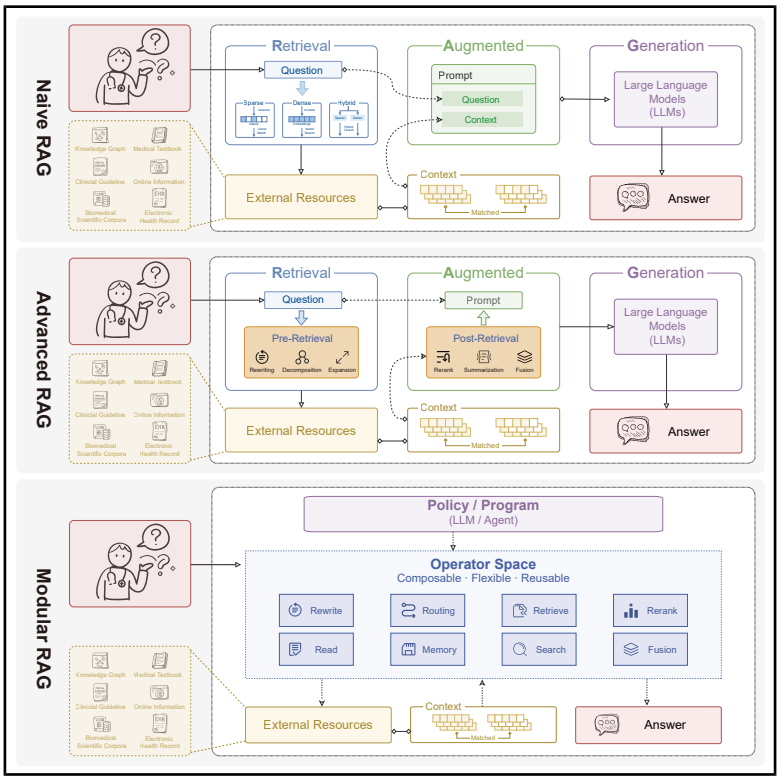
Cell Reports Medicine

Meta-analysis/systematic review

Retrieval-augmented generation in medicine: A scoping review of technical implementations, clinical applications, and ethical considerations

Graphical abstract



Highlights

- A review of 251 medical RAG studies maps technology, clinical use, and ethics
- Private data use remains limited, and English-centric retrieval dominates
- Insufficient evaluation of bias and safety and inadequate attention to equity
- Medical RAG is at an early stage and requires advances to enable responsible global deployment

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In brief

Yang et al. review 251 studies of RAG in medicine, summarizing technical implementations, clinical applications, and ethical considerations. They identify limited evaluation of bias and safety, insufficient attention to equity, and key priorities for enabling responsible global deployment of medical RAG.

Yang et al., 2026, Cell Reports Medicine 7, 102927
August 18, 2026 © 2026 The Author(s). Published by Elsevier Inc.
https://doi.org/10.1016/j.xcrm.2026.102927

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Retrieval-augmented generation in medicine: A scoping review of technical implementations, clinical applications, and ethical considerations

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https://doi.org/10.1016/j.xcrm.2026.102927

SUMMARY

The rapid growth of medical knowledge and the increasing complexity of clinical practice pose challenges. In this context, large language models (LLMs) demonstrate value; however, inherent limitations remain. Retrieval-augmented generation (RAG) shows potential to enhance their clinical applicability. This study reviews RAG applications in medicine. We find that research primarily relies on publicly available data, with limited use of private data. For retrieval, approaches commonly rely on English-centric embedding models, while LLMs are mostly generic, with limited use of medical-specific LLMs. For evaluation, automated metrics evaluate generation quality and task performance, whereas human evaluation focuses on accuracy, completeness, relevance, and fluency, with insufficient attention to bias and safety. RAG applications are concentrated on question answering, report generation, text summarization, and information extraction. Overall, medical RAG remains at an early stage, requiring advances in clinical validation, cross-linguistic adaptation, and support for low-resource settings to enable trustworthy and responsible global use.

INTRODUCTION

Contemporary clinical practice is facing the rapid expansion of medical knowledge and the increasing complexity of diagnostic and therapeutic decision-making.[1] At the same time, the growing demand for personalized health care has introduced unprecedented challenges in the retrieval, integration, and utilization of medical information, further intensifying clinicians' workload.[2] In this context, large language models (LLMs) are gradually being introduced into medicine and have demonstrated potential value. Existing proprietary LLMs[3–13] and open-weight LLMs[14–24] have shown outstanding performance in generative tasks, while reasoning-oriented LLMs have further extended the boundaries of complex tasks.[25–28] Meanwhile, specific LLMs such as Med-Gemini[29,30] and MedGemma[31] have achieved high adaptability to medical scenarios by incorporating specific knowledge.[32] Collectively, these advances highlight the potential applications of LLMs in clinical consultation,[33] disease diagnosis,[34–37] treatment management,[33] medical education,[38] and scientific research,[39] while also suggesting their role in improving overall health care quality.[32,33]

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Despite this, LLMs face significant challenges in medical applications. First, LLMs rely on static training data, limiting their ability to keep pace with rapidly evolving medical knowledge.[40] Second, they are prone to generate content without factual grounding.[36] Third, LLMs typically function as "black boxes" with outputs lacking explainability.[36] Fourth, they cannot access private patient-specific data or hospital-specific guidelines, restricting their utility in personalized diagnostic and treatment support.[40] Lastly, LLMs may perpetuate inherent biases from training data, potentially exacerbating disparities for specific populations and further widening health inequities.[41] These issues are not only technical bottlenecks but also involve safety and ethical governance as well as global health equity.[42]

Against this backdrop, retrieval-augmented generation (RAG) has emerged as a promising solution.[40,43] RAG technology incorporates an explicit, query-driven retrieval process to extract relevant information from external knowledge sources and integrate it into LLMs, ensuring that the outputs are up-to-date, relevant, and fact grounded. As shown in Figure 1, RAG can be categorized into three paradigms according to architectural complexity.[36,44] Naive RAG follows an "index-retrieve-generate" pipeline, in which the system retrieves information from external knowledge sources and augments LLMs with the relevant content to support answer generation.[40] Building on this, advanced RAG introduces pre-retrieval optimization (e.g., query rewriting) and post-retrieval processing (e.g., reranking) to improve retrieval precision and generation quality.[36] Modular RAG goes further by decomposing the fixed pipeline into independently interchangeable functional modules (e.g., a routing module and a memory module) and supports adaptive retrieval strategies such as iterative and recursive retrieval, enabling more tailored configurations for complex medical scenarios. These paradigms represent a progressive evolution from basic to advanced designs, offering potential pathways to mitigate the limitations of LLMs in medical applications.[36,44]

Several studies have examined the development of RAG in medicine, providing valuable perspectives on this rapidly evolving field. Yang et al. analyzed the possible contributions that RAG could bring to health care in equity, reliability, and personalization.[40] Liu et al. conducted a meta-analysis of 20 studies, demonstrating that RAG systems improved the performance of LLMs, and proposed guidelines for the unified implementation and development of enhanced LLM applications with RAG in clinical settings.[45] Amugongo et al. carried out a systematic review with a focus on methodology, analyzing 70 studies to compare different RAG paradigms and their technical implementations in medical scenarios.[46] He et al. surveyed RAG datasets, technologies, and applications in medicine, with particular emphasis on technical components and system architectures.[43] While existing studies offer important insights, they have not systematically mapped the implementation pathways and application patterns of RAG in medicine. More critically, the evaluation of RAG technologies, especially concerning bias, safety, and deployment in low-resource settings, remains insufficiently explored. However, these factors are crucial for advancing the equitable global application of RAG technologies.

This scoping review aims to systematically outline the research landscape of RAG in medicine, mapping its implementation pathways and application patterns and evaluating its potential value in addressing the rapid evolution of medical knowledge and other critical dimensions of clinical practice. Additionally, we emphasize that while RAG alleviates certain limitations of LLMs and facilitates their applications in medicine, it simultaneously introduces challenges. These challenges extend beyond technical reliability and data privacy protection, encompassing human-centered evaluation and oversight, such as bias identification, safety monitoring, and the assurance of cross-linguistic and cross-cultural fairness. Recognizing these issues is crucial for promoting the responsible application of RAG in clinical practice and advancing its role in improving the quality of global health care services.[40]

RESULTS

We retrieved a total of 3,980 study records from PubMed, Embase, Web of Science, and Scopus. After deduplication and screening of titles, abstracts, and full texts, 244 studies met the inclusion criteria. Additionally, 7 relevant studies were manually identified, resulting in a total of 251 studies being analyzed. The PRISMA flow diagram is shown in Figure 2.

Components of RAG framework

External retrieval data

In terms of data source (Figure 3, data source), most studies (80.35%, 184/229) relied on publicly available data, while only 35 studies (15.28%) used private data, and 10 studies (4.37%) used both public and private data. This distribution suggests that research primarily relies on publicly available resources, with limited use of proprietary clinical data.

Meanwhile, RAG research primarily relies on diverse biomedical and clinical resources. As shown in Figure 3 (data type), biomedical scientific corpora constitute the main data source, used in 88 studies, with PubMed being a typical example. Clinical guidelines follow closely, adopted in 67 studies and underscoring the key role of evidence-based medicine in RAG applications. Online information (42 studies) and electronic health records (EHRs) (33 studies) are also common; the former reflects the demand for continuously updated medical knowledge, while the latter represents data support closely tied to real clinical practice. In addition, medical textbooks are incorporated in 33 studies as foundational learning and reference materials; knowledge graphs (30 studies), as a structured form of knowledge representation, offer different possibilities for medical knowledge retrieval; and custom-built datasets (30 studies) reflect researchers' exploration of constructing tailored resources for specific contexts. Overall, the data types involved in medical RAG encompass both structured and unstructured information, forming a diversified support framework.

Retrieval method

As shown in Figure 3 (retrieval method), dense retrieval dominated among the included studies, with 188 studies (83.93%) adopting this approach. In contrast, sparse retrieval was used in only 12 studies (5.36%), while hybrid approaches were applied in 24 studies (10.71%). Dense retrieval methods were primarily implemented in two ways: one relied on general embedding models, such as OpenAI's text-embedding series[47]; the other employed medical-specific embedding models including

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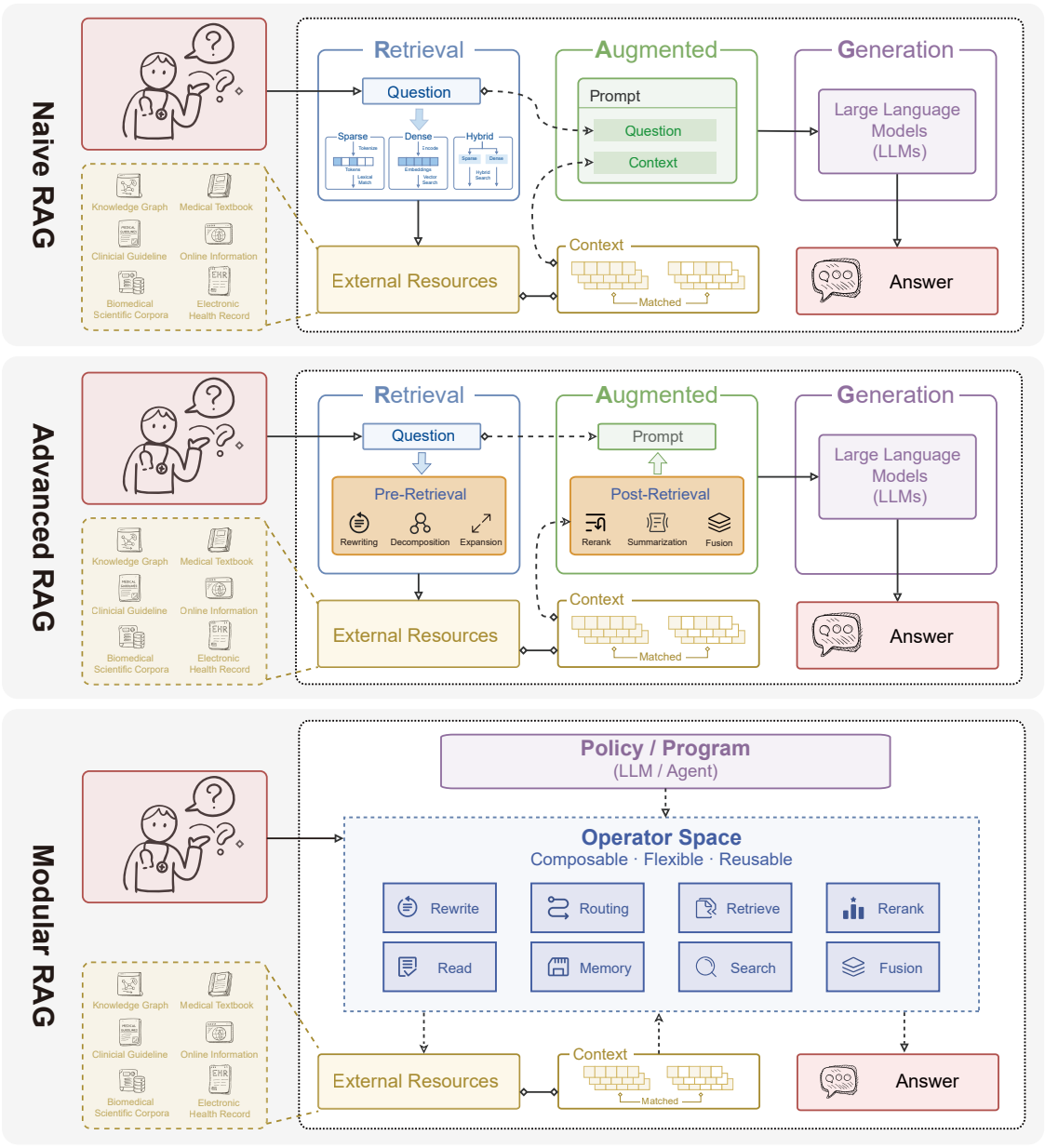


Figure 1. Overview of naive, advanced, and modular RAG frameworks

The naive RAG follows a standard "index-retrieve-generate" pipeline, where relevant information is retrieved and provided to the LLM to generate responses. Advanced RAG extends this pipeline by incorporating pre-retrieval and post-retrieval optimization strategies to improve retrieval precision and generation quality. Modular RAG further introduces flexible functional modules, enabling customized configurations for complex medical scenarios.

BioBERT[48] and MedCPT,[49] which are trained on biomedical corpora to better capture the semantics of the field. However, it is noteworthy that among the 188 studies that applied dense retrieval, only 24 explicitly employed multilingual embedding models (e.g., text-embedding-3[50] and bge-m3[51]), while the remaining studies relied on English-centric embedding models. This is evident when handling multilingual medical data, localized clinical guidelines, and non-English patient records, where semantic misinterpretation and reduced retrieval accuracy may occur.[40,52] Sparse retrieval methods mainly relied on the traditional BM25 algorithm,[53] but their usage frequency within the RAG framework is noticeably low. Hybrid retrieval approaches, which combine sparse and dense methods to balance lexical matching and semantic representation, exhibited certain advantages in some studies.[54]

Generative LLM

As shown in Figure 3 (LLM type), proprietary LLMs were the most widely used, appearing in 100 studies (41.15%). Open-weight LLMs were utilized in 86 studies (35.39%) and a combination of both in 57 studies (23.46%). Proprietary LLMs were primarily

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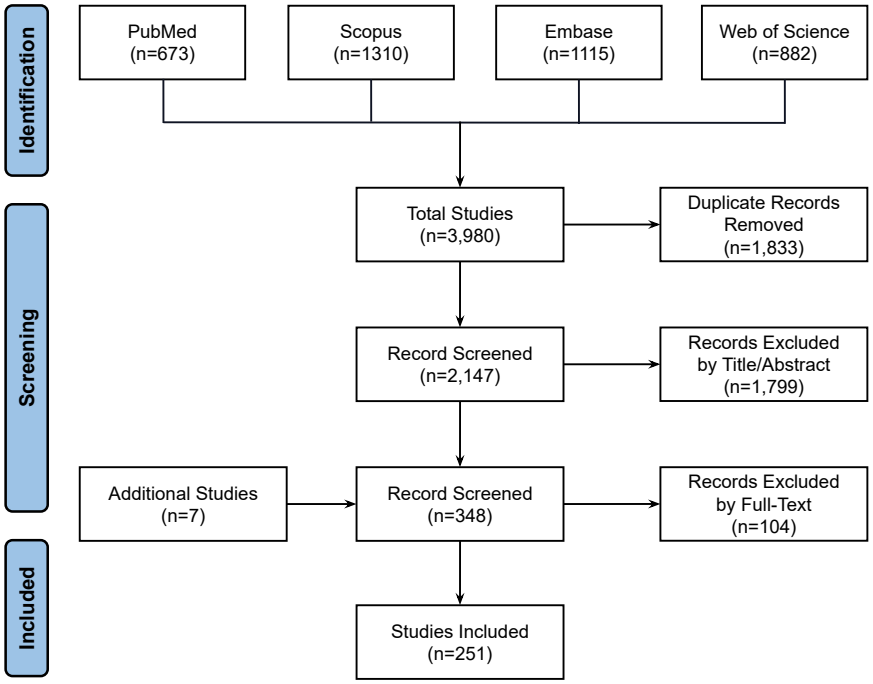


Figure 2. PRISMA flow diagram for identifying related studies

Our search retrieved 3,980 study records (n = 673, 16.91% from PubMed; n = 1,310, 32.91% from Scopus; n = 1,115, 28.02% from Embase; n = 882, 22.16% from Web of Science); of these, 2,147 (53.94%) were retained after deduplication. Following title, abstract, and full-text screening, 244 studies were included, along with 7 additional studies manually identified as relevant, resulting in a total of 251 included studies.

from the OpenAI's GPT series,[10,11,55] while open-weight LLMs mainly include the DeepSeek, Gemma, LLaMA, and Qwen series.[14–25] Notably, medical-specific LLMs were rarely applied in existing RAG research. In this study, medical-specific LLMs refer to models pre-trained on large-scale medical corpora or fine-tuned on medical instruction data. One possible reason is that proprietary medical LLMs (e.g., Med-Gemini[29,30]) have not released APIs to the public, limiting their accessibility for research and practice. Additionally, open-weight medical LLMs have developed more slowly than general LLMs, lagging in both scale and performance.[56]

Medical specialty and application scenario

Medical specialty distribution

The distribution of RAG across medical specialties exhibits variability. Internal medicine occupies the most prominent position, with 45 studies, reflecting its broad demand for synthesizing extensive clinical knowledge. Beyond this, psychiatry and neurology (24 studies), along with radiology (22 studies), have shown a certain level of research activity. Additional specialties, including preventive medicine, emergency medicine, orthopedic surgery, medical genetics and genomics, and obstetrics and gynecology, have been explored to a lesser extent, while engagement in the remaining specialties is limited.

Clinical application scenario

Meanwhile, RAG has been explored in a variety of medical scenarios. Among them, medical question answering (QA) is the most widely used task, which supports clinicians with evidence retrieval, diagnostic reasoning, and clinical decision-making, while also assisting patients in obtaining understandable answers to medical inquiries.[57] Report generation is another application, where structured or semi-structured clinical data are automatically transformed into complete clinical reports (e.g., radiology or pathology reports) to reduce the documentation burden.[57] Additionally, text summarization and information extraction have gained attention: the former condenses lengthy narratives to enhance information accessibility, while the latter converts unstructured text into structured data to support downstream analysis.[57] Other tasks, such as text simplification, have also been explored. Collectively, these tasks demonstrate the multifaceted potential of RAG in supporting clinical workflows.

Evaluation and ethical considerations

Automatic and human evaluation

In medical applications of RAG, evaluation methods demonstrate a relatively balanced distribution between automated and human evaluations, as shown in Figure 4 (automatic and human evaluation). Automated evaluation was the most common setting, adopted in 115 studies (48.94%), while 41 studies (17.45%) only relied on human evaluation, and 79 studies (33.62%) combined both evaluations. Automated metrics generally fall into two categories: (1) text generation quality metrics (e.g., ROUGE,[58] BERTScore,[59] BLEU,[60] and METEOR[61]), which assess linguistic quality and semantic similarity between generated content and reference texts; and (2) task-specific performance metrics (e.g., accuracy, recall, F1 score, and AUROC), typically applied in tasks such as multiple-choice QA and clinical risk prediction. Human evaluation, by contrast, plays an indispensable role in dimensions not fully captured by automated metrics, with a primary focus on the factual accuracy and clinical utility of generated content, including completeness, relevance, fluency, as well as the detection of hallucination, bias, and safety concerns. Overall, current evaluation practices indicate the importance of balancing quantifiable performance metrics with subjective judgments of clinical acceptability.

Ethical and contextual consideration

As shown in Figure 4 (ethical and contextual consideration), only 7 (2.79%) studies explicitly evaluated bias, 24 (9.56%) studies addressed safety, and 6 (2.39%) studies focused on low-resource settings. In terms of bias evaluation, existing studies primarily concentrated on demographic and socioeconomic biases (e.g., age, gender, race, ethnicity, education level, and

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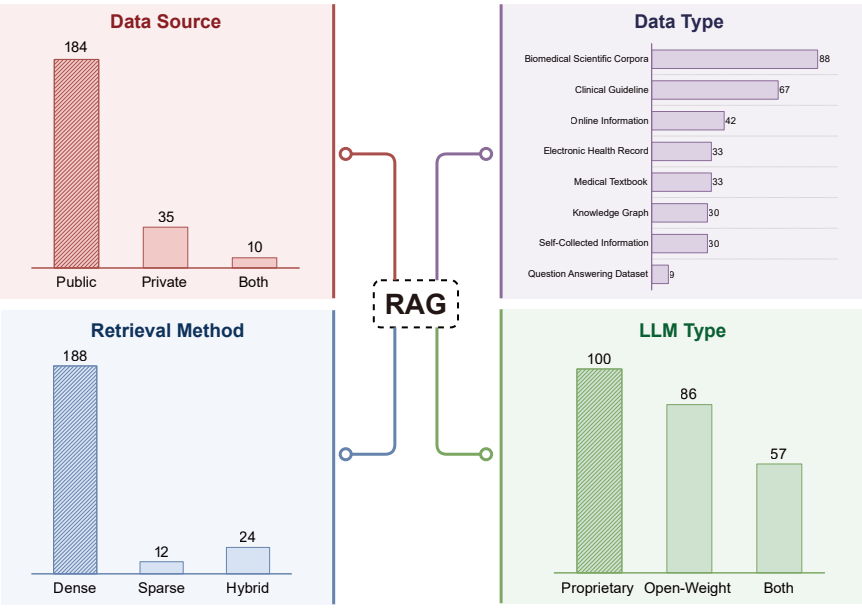


Figure 3. Distribution of external retrieval data, retrieval methods, and generative LLMs among the 251 studies

In terms of data source, 184 studies (80.35%) used public data, 35 studies (15.28%) used private data, and 10 studies (4.37%) used both.

In terms of data type, the studies mainly employed biomedical scientific corpora (88 studies), clinical guideline (67 studies), online information (42 studies), electronic health record (33 studies), medical textbook (33 studies), knowledge graph (30 studies), self-collected information (30 studies), and question answering dataset (9 studies).

As for retrieval method, 188 studies (83.93%) applied dense retrieval, 12 studies (5.36%) applied sparse retrieval, and 24 studies (10.71%) applied hybrid retrieval.

Regarding LLM type, 100 studies (41.15%) applied proprietary LLMs, 86 studies (35.39%) applied open-weight LLMs, and 57 studies (23.46%) applied both.

It should be noted that only information explicitly reported in the studies is included here, while some studies did not provide detailed descriptions.

Information that was not explicitly reported accounted for 22 studies (8.76%) for data source, 4 studies (1.59%) for data type, 27 studies (10.76%) for retrieval method, and 8 studies (3.19%) for LLM type.

socioeconomic status). One study assessed bias from the perspective of factual consistency, while the remaining studies did not clearly define bias types but instead treated bias as one dimension within a multi-criteria evaluation framework. Evaluation approaches mainly relied on small-scale expert assessments, such as inviting clinicians to rate bias in LLM outputs, while others directly analyzed differences across different groups. For safety evaluation, the 24 studies can be categorized into five groups: (1) hallucination-related risks, primarily involving factually inconsistent outputs; (2) clinical decision risks, assessing whether LLMs generate recommendations that may harm patients; (3) medication safety, covering issues such as drug interactions, prescribing errors, and inappropriate dosages; (4) adversarial safety, evaluating LLM robustness against malicious prompts; and (5) data privacy and security, including patient data protection and mechanisms for filtering harmful content. In addition, studies concerning low-resource settings involved countries such as India[62,63] but overall remained very limited. These findings point to an imbalance in current evaluation practices, with insufficient attention given to ethical considerations.

DISCUSSION

This scoping review systematically analyzed 251 RAG studies in medicine, revealing multifaceted characteristics and trends. Regarding data utilization, most studies relied on publicly available resources, with biomedical scientific corpora, clinical guidelines, and online information being the most common; real-world clinical data such as EHRs were used only to a limited extent, while some studies also adopted medical textbooks and

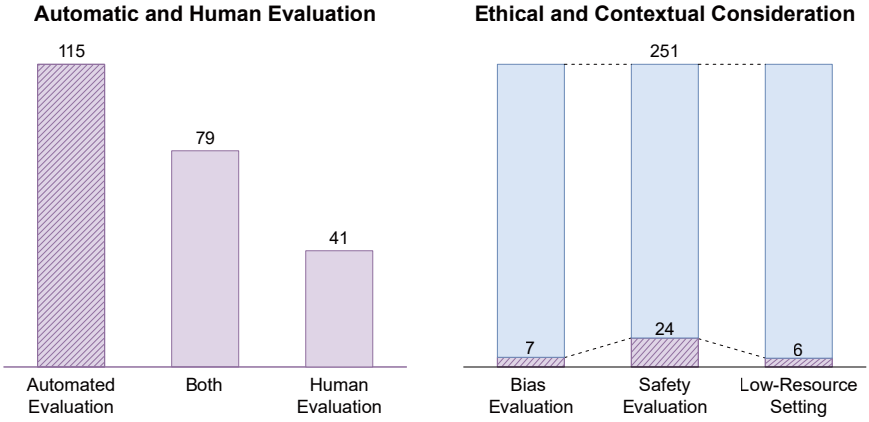


Figure 4. Distribution of evaluation methods and ethical considerations among the 251 studies

A total of 235 included studies reported evaluation methods, with automated evaluation (n = 115, 48.94%), human evaluation (n = 41, 17.45%), and a combination of both (n = 79, 33.62%).

Evaluation methods were not explicitly reported in 16 studies (6.37%). Among all 251 studies, only 7 studies explicitly examined bias, 24 studies addressed safety, and 6 studies were conducted in low-resource settings.

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knowledge graphs. In terms of retrieval methods, dense retrieval emerged as the predominant approach, while sparse and hybrid strategies were reported less frequently. For the generative model, proprietary LLMs dominated, with open-weight LLMs also used to some extent; however, LLMs specialized for medicine were rarely employed. Regarding specialty distribution, the included studies were primarily concentrated in internal medicine, possibly reflecting its reliance on textual longitudinal record integration and multimorbidity reasoning. Other specialties were represented as well, although the overall distribution remained uneven. As for applications, medical QA emerged as the main task. Other important applications include report generation, text summarization, and information extraction, all of which primarily focus on reducing the workload of clinicians. In terms of evaluation methods, automated evaluation was the main strategy, though many studies incorporated human evaluation to compensate for the inadequacies of automated metrics in clinical contexts. Notably, only a small number of studies considered bias, safety, or low-resource settings.

The selection patterns of data sources, to some extent, reveal the constraints faced by the development of medical RAG. The widespread use of publicly available data has facilitated technical validation; however, its content is largely limited to general medical knowledge, making it insufficient to meet the demands of personalized medicine that relies on real clinical data. In contrast, the integration of private data remains limited, primarily constrained by a combination of factors including stringent data privacy and compliance requirements, inadequate cross-institutional collaboration mechanisms, and the complexity of system implementation, all of which constitute key bottlenecks to the translation of RAG technology into clinical settings.[40] To address these challenges, several potential technical pathways can be considered, including localized deployment within health care institutions to complete retrieval and generation processes and thereby reduce the risk of data leakage; the use of federated learning to enable cross-institutional collaborative modeling without sharing raw data; and the introduction of privacy-preserving computing techniques, such as differential privacy and homomorphic encryption, to maintain system performance while ensuring data security.[64]

In terms of retrieval methods, the dominance of dense retrieval in medical RAG applications reflects its advantage in capturing medical semantic understanding, particularly with the use of medical-specific embedding models such as MedCPT,[49] which shows the importance of domain adaptation for retrieval performance. However, current dense retrieval methods exhibit notable limitations at the language level. First, the reliance on English-centric embedding models may hinder effective coverage of non-English medical data, thereby limiting the broader application of RAG technologies in global health care systems.[65,66] Second, such linguistic bias may exacerbate inequities in medical research and clinical practice, especially for communities using underrepresented languages, where relevant medical knowledge is less likely to be adequately incorporated and utilized.[67] Notably, among the 188 studies employing dense retrieval, only 24 (12.77%) explicitly used multilingual embedding models, a relatively low proportion that underscores the insufficient attention given to multilingual retrieval capabilities in current medical RAG research. Although existing multilingual embedding models possess a certain degree of cross-lingual capability, they are not specifically designed for medical contexts, making it difficult to fully capture fine-grained semantic information. Therefore, developing multilingual embedding models tailored to the medical domain is of critical importance for achieving precise semantic representation and high-quality retrieval across multilingual medical texts. At the same time, future studies should strengthen the independent quantitative evaluation of retrieval accuracy and incorporate standalone assessment of the retrieval stage into standard experimental designs, in order to more precisely identify the sources of system errors.

A noteworthy phenomenon in the application of generative LLMs is the relatively low utilization of medical-specific LLMs. Among the 251 included studies, the adoption of such models was extremely limited: on the one hand, proprietary medical LLMs remain insufficiently accessible; on the other hand, open-weight medical LLMs still lag behind general LLMs in terms of overall performance. Consequently, medical RAG research currently continues to favor general LLMs in model selection. Although general models demonstrate strong capabilities, they may still exhibit limitations in handling complex medical terminology, clinical reasoning, and the integration of domain-specific knowledge. Future research should further evaluate the potential value of medical-specific LLMs in particular clinical scenarios.

The distribution of tasks in medical RAG applications reflects differences in both acceptance and practical demand across task types. Among them, QA applications dominate, with their core mechanism being the retrieval of external knowledge to support medical information access, diagnostic reasoning, and clinical decision-making. Based on the nature of these tasks, their potential clinical risks are relatively high: if retrieval sources are opaque, evidence is not verified, or retrieved content contains errors, this may mislead clinical judgment and consequently pose risks to patient safety and trust.[68] In contrast, information-processing tasks (such as report generation, text summarization, and information extraction) involve a lower degree of direct intervention in clinical workflows and therefore carry relatively lower clinical risks. Their primary value lies in reducing documentation burdens and improving information management efficiency, making them a more suitable entry point for the adoption of RAG in clinical settings.

The evaluation methods for current medical RAG applications exhibit a relatively diverse pattern, where the combined use of automated and human evaluation reflects the complexity of medical AI assessment. However, this diversity brings challenges in terms of result comparability and reproducibility to some extent. The widespread use of text generation quality metrics and task-specific performance metrics illustrates the varying evaluation priorities across different medical tasks, yet these metrics often fail to fully capture the practical utility and potential risks of models in real-world clinical settings. Although human evaluation can compensate for the limitations of automated metrics to some degree, it relies heavily on expert judgment, is costly, and difficult to scale, thereby constraining its applicability in large-scale research. Meanwhile, existing studies lack systematic validation of the correlation between automated evaluation metrics and human evaluation, which limits the

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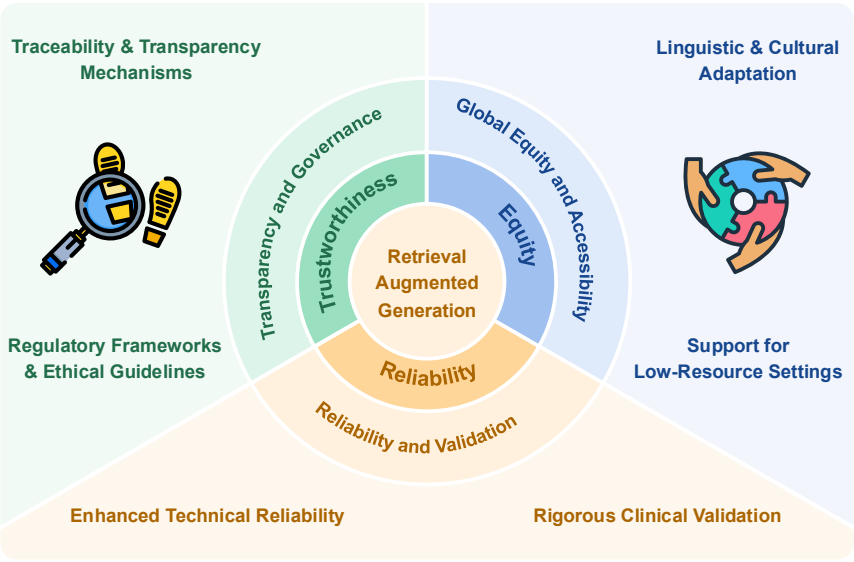


Figure 5. Important directions for advancing medical RAG toward real clinical implementation

This figure outlines three strategic directions required for responsible deployment of RAG in medicine: (1) reliability—ensuring technical robustness and rigorous clinical validation to guarantee system-level stability and clinical actionability; (2) trustworthiness—establishing traceability and transparency mechanisms alongside regulatory frameworks and ethical guidelines; and (3) equity—promoting cross-linguistic and cross-cultural adaptation while supporting low-resource settings to reduce global health disparities.

understanding of the clinical validity of these automated metrics. Notably, only 2.79% of studies explicitly evaluated bias, 9.56% addressed safety, and 2.39% focused on low-resource settings; given the potential clinical implications, these proportions are strikingly low. This limited evidence base makes it difficult to determine whether RAG systems can perform reliably across diverse populations, languages, health care infrastructures, and clinical workflows. Therefore, future studies should more systematically incorporate these evaluations into standard assessment frameworks, rather than treating them as peripheral considerations.

To advance RAG toward real-world clinical implementation, breakthroughs are needed across multiple critical dimensions, as illustrated in Figure 5. First, beyond continuous improvements in technical reliability, more clinical validation is needed, including prospective evaluation in controlled or simulated environments and ultimately application testing within real clinical workflows. Second, it is equally critical to establish traceability and transparency mechanisms that enable clinicians to examine the retrieval sources and reasoning processes underlying generated outputs. In practice, this should be reflected in explicit attribution of retrieved sources, visualization of reasoning chains, and the development of human-AI interaction interfaces that support clinicians in verifying model outputs. Meanwhile, corresponding regulatory frameworks and ethical guidelines must be developed in parallel to ensure patient safety and strengthen trust between health care providers and patients. Finally, it is important to recognize that the global deployment of medical RAG will encounter multidimensional challenges arising from linguistic, cultural, resource, and institutional disparities. Therefore, substantial progress in cross-linguistic and cross-cultural adaptation, as well as the assurance of fairness in low-resource settings, will be required before international implementation.[69] Only by addressing these challenges can RAG achieve safe, trustworthy, and responsible clinical use on a global scale.[70]

Limitations of the study

This study has several limitations. First, we excluded non-English publications, which may introduce bias into the overview of RAG research in medicine. However, our re-examination of four core databases identified only two eligible non-English studies, suggesting that their exclusion had limited impact on the overall conclusions. This pattern reflects the concentration of medical RAG research in English-language publications and may inadvertently reinforce the dominance of English medical knowledge, warranting further attention. Second, AI research is typically published in peer-reviewed conference proceedings, which are not well covered by existing search databases. Although we have manually incorporated representative medical RAG studies from conferences, it remains impossible to capture all relevant conference studies. In addition, given the extremely rapid development of this area, some recent relevant studies were inevitably missed during the search process. Lastly, a proportion of the included studies did not explicitly report key methodological variables. Such incomplete reporting may affect the completeness of data extraction and could introduce structural bias into the overall characterization of this review to some extent.

RESOURCE AVAILABILITY

Lead contact

Requests for resources, data, and materials should be directed to and will be fulfilled by the lead contact, Nan Liu (liu.nan@duke-nus.edu.sg).

Materials availability

This study did not generate new unique materials or reagents.

Data and code availability

- All data reported in this paper will be shared by the lead contact upon request.
- This paper does not report the original code.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

ACKNOWLEDGMENTS

This work was supported by the Duke-NUS Signature Research Program funded by the Ministry of Health, Singapore. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the authors and do not reflect the views of the Ministry of Health.

AUTHOR CONTRIBUTIONS

R.Y. and N.L. contributed to the conceptualization and methodology design. R.Y., M.Y.H.W., H.L., X.L., W.Z., J.L., K.Y., and J.C.K.L. contributed to data curation. R.Y., H.L., Y.C., and W.X. contributed to visualization. R.Y., M.Y.H.W., H.L., and Y.K. drafted the manuscript, with further improvements by J.C.L.O., D.T., C.H., D.S.W.T., and N.L. N.L. supervised the study. All authors contributed to the revision of the manuscript and approved the final version.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

Supplementary data related to this article can be found online at https://doi.org/10.1016/j.xcrm.2026.102927.

Received: November 13, 2025
Revised: April 15, 2026
Accepted: June 24, 2026

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Please cite this article in press as: Yang et al., Retrieval-augmented generation in medicine: A scoping review of technical implementations, clinical applications, and ethical considerations, Cell Reports Medicine (2026), https://doi.org/10.1016/j.xcrm.2026.102927

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Please cite this article in press as: Yang et al., Retrieval-augmented generation in medicine: A scoping review of technical implementations, clinical applications, and ethical considerations, Cell Reports Medicine (2026), https://doi.org/10.1016/j.xcrm.2026.102927

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
PubMed	NCBI	https://pubmed.ncbi.nlm.nih.gov
Embase	Elsevier	https://www.embase.com
Scopus	Elsevier	https://www.scopus.com
Web of Science	Clarivate	https://www.webofscience.com
Software and algorithms		
Microsoft Excel	Microsoft Corporation	https://www.microsoft.com/microsoft-365/excel
Rayyan	Rayyan	https://www.rayyan.ai

EXPERIMENTAL MODEL AND SUBJECT PARTICIPANT DETAILS

This study did not involve experimental models or human subjects.

METHOD DETAILS

Search strategy

We conducted this scoping review following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension for Scoping Reviews (PRISMA-ScR) guideline to ensure transparency and rigor throughout the process.[71] We performed a comprehensive search across PubMed, Embase, Web of Science, and Scopus. The search strategy involved using a combination of terms related to "RAG" and "Medicine" to ensure maximum coverage and relevance. Our search period spans from 2017 (when the "Transformer" architecture was introduced[72]) to July 1st, 2025, to ensure comprehensive coverage of the full development trajectory of RAG technologies in the context of LLMs. Additionally, during the revision process, we manually incorporated selected recent studies to improve the timeliness of this scoping review; however, this supplementary update did not constitute a complete rerun of the systematic database search.

Inclusion criteria and exclusion criteria

Studies were included if they met the following criteria: (1) Implemented a RAG framework, defined as incorporating an explicit, query-driven retrieval process to extract relevant information from external knowledge sources and integrate it into LLMs to improve generation; and (2) Were applied in medical scenarios, with potential impact and contributions to health care services. Additionally, we excluded studies that lacked abstracts, were non-English, were not peer-reviewed, were non-research types (such as review or perspective), as well as studies without clear RAG implementation detail.

In the title and abstract screening stage, each study was independently screened by at least two researchers (RY, MW, HL, XL, WZ, JL, KY and JL) to determine eligibility. During the full-text screening stage, at least two researchers again independently screened the studies and extracted data. Any disagreements arising during this process were resolved through consultation with domain experts (RY, WX).

Data extraction

For each included study, we extracted information across the following 6 perspectives: (1) metadata; (2) external retrieval data: data source, data availability, data scale, retrieval method; (3) model: generative model, model accessibility, embedding model; (4) application: medical specialty, application scenario; (5) evaluation: evaluation metrics, inclusion of human evaluation; (6) bias, safety, and low-resource setting: inclusion of bias evaluation, inclusion of safety evaluation, and application in low-resource settings, defined as health care contexts in low- and middle-income countries, where medical resources and infrastructure are relatively limited.

QUANTIFICATION AND STATISTICAL ANALYSIS

No statistical analysis or quantification was performed beyond descriptive counts and proportions as reported in the Results.

ADDITIONAL RESOURCES

No additional resources were generated for this study.